

VOPE: Revisiting Hallucination of Vision-Language Models in Voluntary Imagination Task

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Abstract

Most research on hallucinations in Large Vision-Language Models (LVLMs) focuses on factual description tasks that prohibit any output absent from the image. However, little attention has been paid to hallucinations in voluntary imagination tasks, such as story writing, despite this human-like cognitive ability being essential for real-world generative applications. To address this limitation, we introduce **Voluntary-imagined Object Presence Evaluation (VOPE¹)**—a recheck-based evaluation benchmark for assessing LVLMs’ grounding behavior in voluntary imagination tasks. Specifically, VOPE poses recheck-based questions to evaluate how an LVLM interprets the presence of the imagined objects in its own response. Rather than penalizing the imagined content itself, VOPE identifies hallucinations based on the correctness of the model’s presence judgments for the generated objects. Built on this idea, we construct a dataset covering captioning, reasoning, and writing tasks with different levels of voluntary imagination. We apply VOPE to several mainstream LVLMs and hallucination mitigation methods, revealing two key findings: (1) most LVLMs hallucinate heavily during voluntary imagination, and their performance in presence evaluation is notably poor on imagined objects; (2) existing hallucination mitigation methods show limited effect in voluntary imagination tasks, making this an important direction for future research.

CCS Concepts

• **Computing methodologies** → **Computer vision**; **Natural language processing**.

Keywords

LVLM hallucination, voluntary imagination, multimodal evaluation

1 Introduction

Recent advancements in Large Language Models (LLMs) have led to performance that matches or surpasses human-level performance

across various text-based tasks [1, 9, 14, 44, 46]. By integrating visual encoders and visual instruction tuning, these foundational LLMs evolve into Large Vision-Language Models (LVLMs) that excel at various multimodal tasks [2, 8, 28, 60]. However, previous studies reveal that such models occasionally generate outputs that do not align with the given image input [29]. This phenomenon, often termed hallucination, has been attributed to factors such as the model’s insufficient visual encoding capability and its reliance on the LLM priors during inference [29]. Significant efforts have been dedicated to exploring the origins of hallucinations [15, 17, 24, 30, 49] and developing mitigation strategies [2, 18, 23, 27, 31, 55].

Previous studies on LVLM hallucinations focus primarily on tasks that require factual output, such as image captioning and visual question answering. However, in many real-world applications, such as story writing, LVLMs need to generate novel content or make inferences beyond the given image [11, 35]. This behavior, known as **voluntary imagination** in psychology [26, 47], is an important ability possessed by humans. Unlike hallucination, correct and reasonable imagination can help produce higher-quality outputs in generative tasks. Therefore, existing response-based hallucination evaluations, which strictly demand image-text alignment, become unreliable when they are directly applied to open-ended generation as they tend to penalize valid imagination. A new evaluation framework is needed to assess LVLMs’ hallucinations under such tasks that involve voluntary imagination.

To address this problem, we introduce Voluntary-imagined Object Presence Evaluation (VOPE), a benchmark designed to assess LVLMs’ hallucinations during tasks involving voluntary imagination. As shown in Fig. 1, unlike existing response-based assessments that determine hallucinations based solely on whether the output object exists in the image, VOPE rechecks the model’s interpretation of its own output. The consistency between this interpretation and the object’s presence in the image is used to determine whether a hallucination occurs. The example in Fig. 1 indicates that even when the output object is absent from the image, this should not be considered a hallucination if the model also interprets the object as

¹Our benchmark is released under CC BY-NC 4.0 license. The evaluation data, code, and appendix are available at <https://github.com/qqw5ad5/VOPE>.

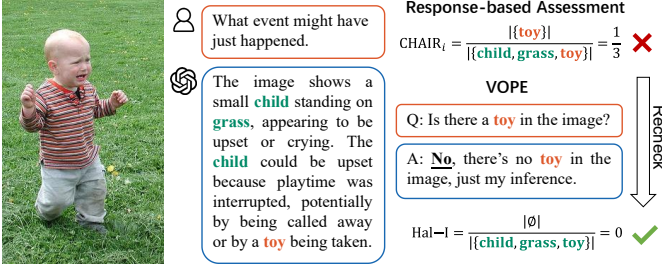


Figure 1: Comparison between VOPE and response-based assessments. In this example, the model outputs an absent object “toy” when inferring about what might have just happened in the image. A response-based assessment such as $CHAIR_i$ would regard “toy” as a hallucination. In contrast, VOPE rechecks the model’s interpretation of the output, identifying “toy” as non-hallucinatory voluntary imagination and correcting the calculation of the hallucination rate.

non-existent. Such a correct interpretation indicates that the model is engaging in non-hallucinatory imagination.

Based on the recheck evaluation, VOPE introduces *Hal-D* and *Hal-I* to quantify hallucination rates in factual description and voluntary imagination, respectively. Experiments reveal that while mainstream LVLMs achieve low *Hal-D*, they suffer from high *Hal-I*, which existing mitigation methods fail to resolve. Additionally, to better analyze the models’ generation preferences in voluntary imagination tasks, we introduce the *Exp* metric to measure their imaginative propensity alongside a relevance assessment to evaluate the contextual appropriateness of the imagined content.

The main contributions of this work are summarized as follows:

- We highlight that generating novel content and inferring beyond the image are crucial for LVLMs in tasks like story writing, suggesting that treating all such voluntary imaginations as hallucinations is inappropriate.
- We introduce a Voluntary-imagined Object Presence Evaluation (VOPE) benchmark for assessing LVLMs under tasks involving different levels of voluntary imagination. VOPE separates factual description from voluntary imagination through recheck-based presence evaluation, enabling a grounding-focused diagnosis beyond response-only assessment.
- We uncover two key findings with VOPE: (1) most LVLMs perform notably poorly when determining the presence of objects they have voluntarily imagined during the generative task; (2) current hallucination mitigation methods show limited efficacy in voluntary imagination tasks, highlighting an important avenue for future work.

2 Related Work

2.1 Hallucination Evaluation on LVLML

In recent years, as Large Vision-Language Models (LVLMLs) have made significant advances, the issue of hallucinations in their outputs has garnered increasing attention [29]. A growing body of research is now dedicated to building benchmarks for evaluating potential hallucination problems in LVLMLs.

One of the most direct approaches involves constructing Visual Question Answering (VQA) tasks that query the existence of objects

in images [17, 24, 34]. However, such evaluation methods are limited in their scope for assessing hallucinations, as their questions for the presence evaluation of each image are fixed.

In parallel, other works focus on hallucinations in LVLMLs’ generative tasks, such as image captioning, by comparing the overlap between objects mentioned in the model’s output and the ground truth objects in the image [39, 49]. However, accurately evaluating whether the content generated by the model aligns with the image content remains a challenging problem. To solve the problem, some researchers employ LLM judges to assist in detecting hallucinated information within the model’s response [16, 21, 40]. While promising, the reliability of these LLM judges is still an area requiring further validation.

Recently, research has expanded to explore fine-grained hallucination categories [19, 53] and their specific causes, including visual understanding failures [15], counter-common-sense content [30], and synthetic image biases [13]. Alongside studies comparing generative and discriminative task performance [51], new evaluation paradigms have emerged. For instance, THRONE [22] introduces a post-generation verification approach; however, it relies on an external judge to verify output factuality. In contrast, VOPE uniquely evaluates whether the target LVLML itself can correctly recognize the presence status of its own generated objects.

2.2 Hallucination Mitigation on LVLML

Given the significant challenges posed by hallucinations in LVLMLs, numerous studies have focused on developing methods to mitigate or eliminate these issues. These mitigation approaches can be broadly categorized into two groups: those applied during the training phase and those applied during the inference phase.

In the training phase, mitigation methods aim to produce a more robust model that minimizes hallucinations during subsequent use. These strategies include enhancing the capabilities of the visual encoder [2, 27], improving the quality of data in visual instruction tuning [2, 16, 17, 27], and adjusting the model’s output preferences [32, 37, 55].

During the inference phase, hallucinations are reduced by modifying the model’s decoding procedure. One of the most widely used methods is contrastive decoding, which amplifies the difference in output probabilities between positive and negative examples to strengthen the likelihood of non-hallucinatory outputs [12, 23, 36, 41, 48, 59]. One of the most representative works, VCD [23], compares the outputs obtained with original and distorted visual inputs, ensuring that the generated content is closely grounded to the visual inputs. Another approach is model intervention [10, 31, 38], which targets the model’s internal activations associated with hallucinatory outputs, intervening in the inference process to suppress hallucinations. Additionally, some studies focus on the model’s attention mechanism [18, 20, 42, 57], ensuring that attention is properly focused on the correct image content during generation. One unique work, ICT [4], incorporates intervention methods for attention adjustment, shifting the model’s focus to different levels of visual information and enhancing its attention to visual details. Other approaches include feature decoupling [54], retrospective resampling [52], and the introduction of visual grounding [58], all of which aim to further eliminate hallucinations in LVLMLs.

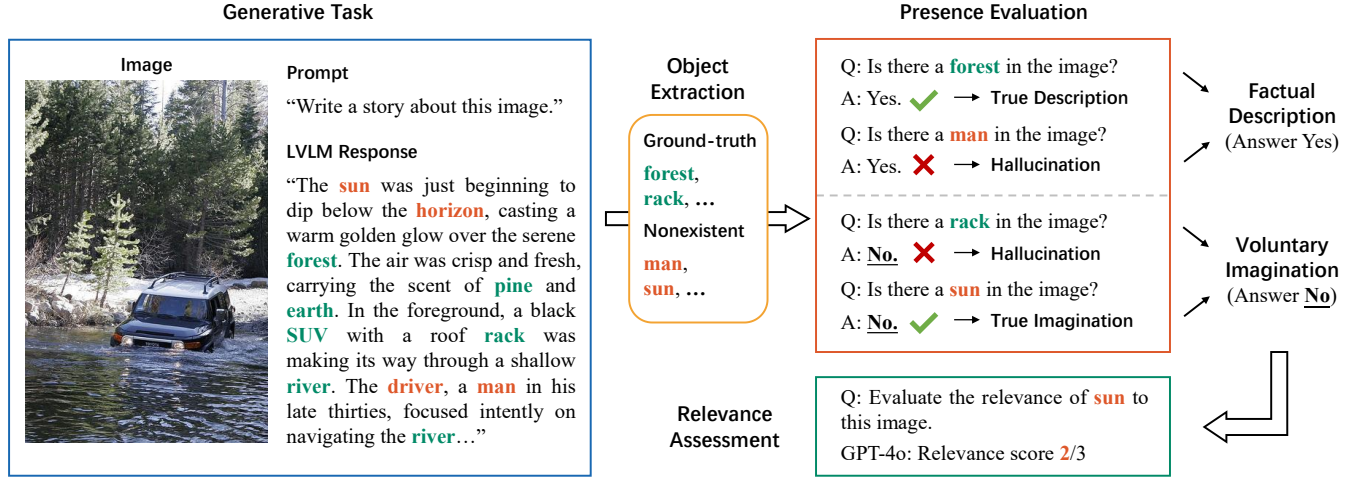


Figure 2: Overview of the Voluntary-imagined Object Presence Evaluation (VOPE). After generating a response based on the prompt, the model is required to evaluate the presence of the objects in the response. All output objects are then categorized into two sets: **Factual Description** (objects that the model believes to exist and outputs accordingly) and **Voluntary Imagination** (objects that the model outputs even though it believes they do not exist in the image). We separately assess hallucination within these two sets. Finally, a relevance check identifies models with excessive irrelevant imagination.

2.3 Creativity Evaluation on LVLm

Some recent benchmarks [11, 33] evaluate whether LVLms can produce creative and context-appropriate responses in creative generation tasks. These works focus on creativity quality, such as novelty and expressiveness. In contrast, our work focuses on hallucination and grounding behavior under voluntary imagination. Therefore, VOPE is not intended to be a general creativity benchmark, but a grounding-focused evaluation benchmark for voluntary imaginations.

3 VOPE Benchmark

3.1 Overview

An overview of the proposed Voluntary-Imagined Object Presence Evaluation (VOPE) is presented in Fig. 2. Unlike existing evaluation methods that assess a model’s hallucination based solely on the consistency between the model’s response and the image content, our approach focuses more on evaluating the correctness of the model’s interpretation of the presence of its output objects. In this sense, VOPE serves as a recheck-based benchmark for diagnosing grounding behavior on generated objects, especially in voluntary imagination settings. In the following sections, we provide a detailed introduction to each step of the VOPE and the metrics employed.

3.2 Presence Evaluation

After the LVLm generates a response to the task, we extract the mentioned objects and ask the model to assess their presence in the image. Based on the model’s answer in the presence evaluation, we compare the consistency between the answers and the ground truth existence labels of these objects. Any inconsistencies in the answers indicate hallucinations in the LVLm.

We further categorize the objects extracted from the generative task response into two types based on the model’s answer in the

presence evaluation: **Factual Description**, which refers to objects the model believes exist in the image and outputs accordingly, and **Voluntary Imagination**, which includes objects the model generates but acknowledges as not existing in the image. We then analyze the correct and hallucinated objects separately within these two categories.

In the case of factual description, a subject widely studied in previous research, the definitions of correct and hallucinated outputs are as follows:

- **True Description:** Objects present in the image that the model correctly identifies. In this case, the model accurately describes the image content.
- **Hallucination:** Objects the model asserts are present in the image but are actually absent. This category aligns with the primary focus of existing LVLm hallucination research.

For voluntary imagination, an important but less-explored aspect, we define true imagination and the corresponding hallucination as follows:

- **True Imagination:** Objects not present in the image, which the model correctly recognizes as absent. These objects are generated as part of voluntary imagination to fulfill task requirements (e.g., story writing). *Previous evaluation methods would mistakenly treat all of these imagined outputs as hallucinations.*
- **Hallucination:** Objects that are actually present in the image but are judged as absent by the model in the recheck stage. They reflect incorrect grounding on the generated objects, and in this work we treat them as a hallucination manifestation under voluntary imagination. *Since these objects are present in both the image and the original response, previous evaluation methods would simply categorize them as correct outputs.*

Table 1: Performance of mainstream LVLMs on VOPE within three generative tasks.

Method	Captioning			Reasoning			Writing		
	Hal-D ↓	Hal-I ↓	Exp	Hal-D ↓	Hal-I ↓	Exp	Hal-D ↓	Hal-I ↓	Exp
LLaVA1.5 [27]	21.2	19.3	13.6	16.9	19.4	12.9	21.9	18.8	14.8
LLaMA3.2-Vision [14]	14.1	23.4	3.7	12.5	21.2	5.3	13.9	17.6	6.0
InternVL2.5 [5]	11.4	37.9	12.6	7.9	38.2	14.0	12.3	29.1	28.0
InternVL3.5 [50]	11.8	38.8	13.1	8.7	34.5	14.1	12.0	28.3	29.3
Qwen2.5-VL [45]	7.3	43.6	15.2	5.2	32.8	23.4	7.3	35.8	30.2
Qwen3-VL [3]	9.0	36.6	16.0	8.0	28.7	21.6	9.1	30.5	36.6
MiniCPM-o 2.6 [56]	9.4	27.6	9.0	7.8	23.0	11.7	10.9	23.6	18.0
Gemma3 [43]	11.9	17.4	5.5	11.3	15.8	7.5	15.2	17.8	24.0

3.3 Metrics

After identifying the hallucinated objects in both the model’s factual description and voluntary imagination output, we propose two metrics, *Hal-D* and *Hal-I*, to evaluate LVLMs’ hallucination rate based on the proportion of the hallucinated objects. For clarity, we define the sets of each category of objects as follows: true description is denoted as $\mathbf{D}_T$, hallucination in factual description is denoted as $\mathbf{D}_H$, true imagination is denoted as $\mathbf{I}_T$ and the grounding error in voluntary imagination is denoted as $\mathbf{I}_H$.

The first metric evaluates the model’s hallucination rate within its factual description:

$$Hal-D = \frac{|\mathbf{D}_H|}{|\mathbf{D}_T| + |\mathbf{D}_H|}. \quad (1)$$

The second metric evaluates the model’s grounding failure rate on voluntarily imagined objects. To maintain consistency with the first metric, we denote it as the hallucination rate under voluntary imagination:

$$Hal-I = \frac{|\mathbf{I}_H|}{|\mathbf{I}_T| + |\mathbf{I}_H|}. \quad (2)$$

Considering that the amount of voluntary imagination may affect the calculation of *Hal-I* metric, we introduce an additional metric to represent the proportion of voluntarily imagined content among the model’s outputs:

$$Exp = \frac{|\mathbf{I}_T| + |\mathbf{I}_H|}{|\mathbf{D}_T| + |\mathbf{D}_H| + |\mathbf{I}_T| + |\mathbf{I}_H|}. \quad (3)$$

3.4 Task Construction

Under the VOPE benchmark, we construct a benchmark based on 5000 samples selected from the MSCOCO dataset [25]. For each sample, we use different prompts to instantiate three generative tasks with different levels of voluntary imagination.

- **Captioning:** LVLM only needs to perform factual descriptions without voluntary imagination.
- **Reasoning:** LVLM engages in some degree of voluntary imagination based on the image content.
- **Writing:** LVLM typically involves substantial voluntary imagination in their response.

Together, these tasks provide a compact setting that spans from factual description to voluntary imagination.

3.5 Relevance Assessment

Furthermore, we observe that excessive, image-irrelevant voluntary imagination can degrade output quality while biasing the value of *Hal-I*. To guard against this, we incorporate a relevance assessment into the VOPE pipeline. We adopt an MLLM-as-a-judge approach, using the latest GPT-4o model to evaluate the relevance of the model-generated objects to the image. For each object within the model’s response that is not present in the image, we ask the judging model to assign a relevance score based on the following criteria:

- **Score 3:** Highly relevant to the image, likely to appear in similar scenarios.
- **Score 2:** Moderately relevant to the image, though its presence does not feel out of place.
- **Score 1:** Completely irrelevant to the image, appearing discordant in the image.

The details of the evaluation prompt are provided in Appendix C.

4 Experiments

4.1 Hallucination Evaluation

We report the evaluation results of eight representative LVLMs on the VOPE benchmark in Table 1. From the *Hal-D* columns in Table 1, most LVLMs exhibit relatively low hallucination rates during factual description, with stronger models such as Qwen2.5-VL and Qwen3-VL achieving the lowest values. However, the *Hal-I* results in the same table indicate that most LVLMs still suffer from high hallucination rates during voluntary imagination. This contrast shows that strong factual-description performance does not guarantee that the model can correctly recognize the presence status of the objects it generates in voluntary imagination. Instead, VOPE reveals a grounding inconsistency that is not directly visible from the response alone.

Qwen2.5-VL is a representative case: it achieves very low *Hal-D* values across tasks, but yields the highest *Hal-I* in both captioning and writing. This suggests that evaluations centered on factual description alone may miss an important failure mode in voluntary imagination. We illustrate a representative example of this grounding inconsistency in Appendix A.

Furthermore, we present an additional attention-based analysis in Appendix B showing that the presence judgments in VOPE

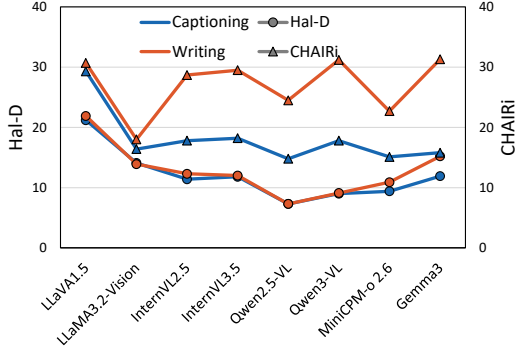


Figure 3: Comparison of the $CHAIR_i$ and $Hal-D$ metrics. $CHAIR_i$ metric overestimates the hallucinations of LVLMs in the writing task.

Table 2: Comparison between $Hal-I$ on the writing task and existing hallucination benchmarks. (Note: POPE* denotes 1 – Acc to align with the $Hal-I$ metric)

Model	$Hal-I$ ↓	POPE* ↓	MMHal ↑	PhD ↑
LLaVA1.5 [27]	18.8	22.1	2.1	32.2
LLaMA3.2-Vision [14]	17.6	14.8	3.8	57.8
InternVL2.5 [5]	29.1	17.7	3.9	59.9
InternVL3.5 [50]	28.3	21.6	3.7	58.7
Qwen2.5-VL [45]	35.8	12.2	3.7	67.6
Qwen3-VL [3]	30.5	14.5	4.8	76.0
MiniCPM-o 2.6 [56]	23.6	15.7	4.2	59.4
Gemma3 [43]	17.8	15.4	3.6	68.4

remain connected to the original generation process, rather than simply reflecting post-hoc self-correction.

4.2 Comparison with Existing Hallucination Benchmarks

We compare VOPE with representative existing hallucination benchmarks to examine whether it captures a different aspect of LVLM behavior.

We first compare the proposed $Hal-D$ metric with the traditional $CHAIR_i$ metric used for evaluating hallucinations in generative tasks. As shown in Fig. 3, in the captioning task, both metrics yield similar evaluation results. This is expected because voluntary imagination is limited in this setting. However, in the writing task, the $CHAIR_i$ metric becomes unreliable and attains spuriously high values. A main reason is that $CHAIR_i$ counts all image-absent outputs from true imagination as hallucinations. In contrast, the proposed $Hal-D$ metric consistently evaluates hallucinations in factual descriptions in such tasks, as it excludes voluntarily imagined objects from the formula. This result suggests that response-based hallucination metrics remain useful in factual generation, but become unreliable once voluntary imagination is involved. We next compare VOPE with broader hallucination benchmarks to examine whether it reveals a different failure pattern.

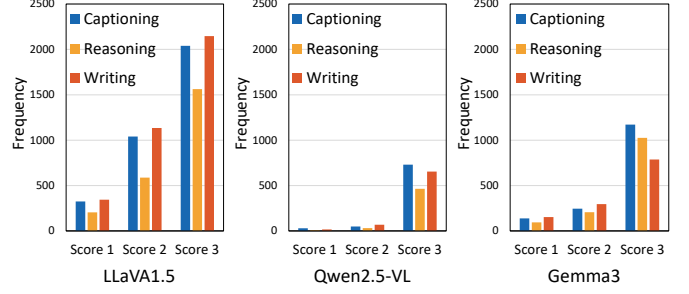


Figure 4: Relevance score distribution for hallucination in factual description. The majority of the hallucinated objects are highly relevant to the input image.

We further compare $Hal-I$ with other existing hallucination benchmarks, including POPE [24], MMHal [40], and PhD [30]. As shown in Table 2, the relative ranking of models differs substantially across these evaluations. This difference is expected because existing benchmarks mainly assess hallucination on pre-defined queries, while VOPE further evaluates the model’s presence judgments regarding its own generated objects. For example, Qwen2.5-VL achieves strong POPE and PhD results, but still yields the worst $Hal-I$ result. Conversely, Gemma3 achieves a high PhD score while maintaining a relatively low $Hal-I$. These differences suggest that VOPE does not merely reproduce the scores of existing hallucination benchmarks. Instead, it captures a distinct and important dimension of LVLM behavior. At the same time, the mismatch between $Hal-I$ and POPE indicates that, although both are discriminative evaluations, $Hal-I$ poses a more demanding test by asking whether the model can correctly recognize the presence status of its own generated objects under voluntary imagination.

4.3 Relevance Assessment

To better understand the nature of absent objects in model outputs, which stem from imagination or hallucination, we further assess their relevance to the image. Representative low- and high-relevance examples are provided in Appendix A, while we focus here on the overall score distributions.

We select three representative models to analyze the distribution of relevance scores in their outputs, as shown in Fig. 4 and Fig. 5. Here, **Score 1**, **Score 2**, and **Score 3** represent objects that are completely irrelevant, moderately relevant, and highly relevant to the image content, respectively. **Frequency** indicates the total number of occurrences of these objects throughout each evaluation task. The complete relevance assessment results for all LVLMs are reported in Appendix E.

From the results in Fig. 4, LLaVA1.5 exhibits a much higher frequency of hallucinations compared to the more powerful Qwen2.5-VL and Gemma3. However, all three models predominantly generate hallucinations that are highly relevant to the image, while irrelevant hallucinations occur only infrequently. This suggests that factual hallucination is often not completely off-image, but instead

Table 3: Effect of hallucination mitigation methods within voluntary imagination tasks.

Method	Captioning			Reasoning			Writing		
	Hal-D ↓	Hal-I ↓	Exp	Hal-D ↓	Hal-I ↓	Exp	Hal-D ↓	Hal-I ↓	Exp
Greedy [28]	19.8 +0.0	17.0 +0.0	7.2 +0.0	14.1 +0.0	12.9 +0.0	9.2 +0.0	19.6 +0.0	13.6 +0.0	8.1 +0.0
DoLA [7]	20.5 +0.6	17.7 +0.7	9.2 +2.0	15.0 +1.0	15.3 +2.4	10.1 +0.9	20.6 +1.0	16.7 +3.1	10.0 +1.9
VCD [23]	17.3 -2.6	20.6 +3.6	9.9 +2.7	13.1 -0.9	18.4 +5.6	9.9 +0.7	17.7 -1.9	19.1 +5.5	11.5 +3.4
OPERA [18]	20.0 +0.1	17.1 +0.1	7.2 +0.0	13.9 -0.2	13.2 +0.3	9.4 +0.2	19.7 +0.1	13.8 +0.2	8.5 +0.4
HALC [6]	17.4 -2.4	17.0 +0.0	7.6 +0.4	12.9 -1.1	14.1 +1.2	9.0 -0.2	17.6 -2.0	15.1 +1.4	8.4 +0.2
VTI [31]	9.3 -10.5	38.9 +21.9	3.1 -4.1	11.5 -2.6	24.2 +11.3	5.9 -3.4	12.1 -7.5	25.5 +11.8	6.4 -1.7

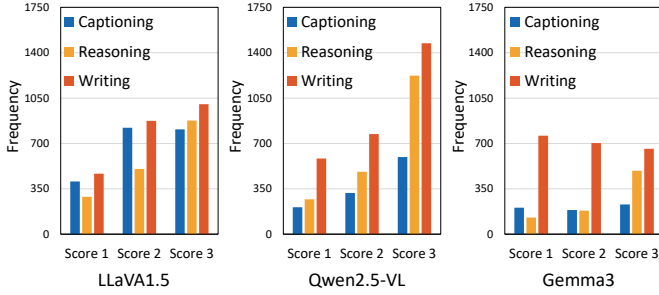


Figure 5: Relevance score distribution for true imagination outputs. While Qwen2.5-VL’s outputs remain highly image-relevant in both reasoning and writing, Gemma3’s writing outputs are notably less tied to the visual context.

Table 4: Agreement results for relevance-score annotation.

Pair	Weighted Kappa
GPT-4o vs Human Group1	0.5326
Human Group1 vs Human Group2	0.6230
GPT-4o vs Gemini	0.6353

reflects a confusion between contextual relevance and actual existence, which is also consistent with previous observations in POPE [24].

For the results with true imagination in Fig. 5, we observe that Qwen2.5-VL outputs the highest number of objects that are highly relevant to the image content. By contrast, the proportion of image-irrelevant objects becomes higher than in factual hallucination, especially for Gemma3 in the writing task. These results suggest that different LVLMs vary not only in hallucination rate, but also in how well their imagined content remains relevant to the image.

To validate the reliability of the LLM-based relevance scoring, we compare the evaluations from GPT-4o with human annotations and another LLM judge Gemini in Table 4. GPT-4o achieves a weighted kappa of 0.5326 against Human Group1, which is reasonably close to the human-human agreement of 0.6230. Moreover, GPT-4o and Gemini reach a kappa of 0.6353, demonstrating strong cross-model consistency. These results suggest that GPT-4o provides a reasonable relevance estimator for this analysis.

4.4 Evaluation of Mitigation Methods

Finally, we investigate the effectiveness of existing hallucination mitigation methods for both factual description and voluntary imagination. Since these methods are mainly designed to reduce factual hallucination, we further examine whether their gains can transfer to the VOPE setting. Given that many hallucination mitigation approaches rely on greedy decoding, we use the results of LLaVA1.5 with greedy decoding as our baseline for comparison [6], as shown in Table 3.

We observe that although some of the mitigation methods can reduce hallucinations in factual description, none of them effectively address hallucinations in voluntary imagination. The *Hal-I* metric across all the methods remains unchanged or even increases significantly compared to the baseline. This result reveals that **current mitigation strategies have notable limitations in addressing hallucinations in voluntary imagination tasks**. One possible explanation is that existing mitigation strategies mainly focus on suppressing unsupported objects in the output. However, VOPE further tests whether the model can consistently interpret the presence of its generated objects, where cases of “correct generation but incorrect interpretation” still remain. This suggests that rather than simply suppressing the output of non-existent objects, it is significantly more important in voluntary imagination tasks to ensure the model correctly interprets the visual presence of what it generates. As an additional observation, we find that when contrastive decoding is only used to adjust model expression, the hallucination rates remain largely unchanged (see Appendix D), which points to another related phenomenon worth further investigation.

5 Conclusion

In this paper, we introduce VOPE, a recheck-based benchmark for evaluating LVLMs in voluntary imagination tasks. Rather than relying solely on response-level matching between the output and the image, VOPE evaluates whether a model can correctly interpret the presence of its own generated objects. Our experiments reveal a striking contrast: while mainstream LVLMs achieve relatively low hallucination rates in factual descriptions, they struggle severely under VOPE in voluntary imagination. Furthermore, we find that existing hallucination mitigation methods, despite their efficacy in factual scenarios, fail to generalize to voluntary imagination tasks. These results suggest that conventional evaluation protocols are insufficient to fully characterize model reliability in creative contexts, highlighting a critical need for future LVLMs with stronger grounding capabilities for their generated content.

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A Qualitative Example

A.1 Case of Hallucination in Voluntary Imagination

We show an example of Qwen2.5-VL’s grounding failure in voluntary imagination in Fig. 6. Although the model’s original output contains “books”, it denies the existence of the “book” in the subsequent presence evaluation. Such cases of “output but incorrect interpretation” indicate that even when the model outputs an object that is present in the image, it may still fail to correctly recognize the presence status of that object.

A.2 Case of Relevance Assessment

We provide representative examples of low- and high-relevance outputs used in the relevance assessment. As depicted in the figure, an object such as a “bed”, which is unlikely to be present in the living room scene, receives a lower relevance score, while an object like a “television” receives a higher relevance score.

B Attention Analysis

To examine the internal focus of LVLs during object generation, we analyze the image attention weights of the LLaVA1.5 model. Specifically, we calculate the total proportion of attention allocated to image tokens when generating an object, defining this as the object’s image attention weight. We subsequently compute the average image attention weight for each object in two conditions: when the object is generated as a factual description and when it is generated as a voluntary imagination.

As shown in Fig. 8, the concentration of points in the lower-right region indicates that the model allocates more attention to image tokens during factual descriptions than during voluntary imaginations. Consequently, the model relies heavily on visual evidence for factual outputs, but shifts its weight toward accumulated textual context when generating imagined content. These distinct attention patterns confirm that the presence judgments evaluated by VOPE are deeply rooted in the original generation process, rather than merely reflecting an independent, post-hoc self-correction.

Furthermore, we visualize LLaVA1.5’s attention to image tokens when generating corresponding objects, as shown in Fig. 9. It can be observed that the patterns of attention maps differ between true

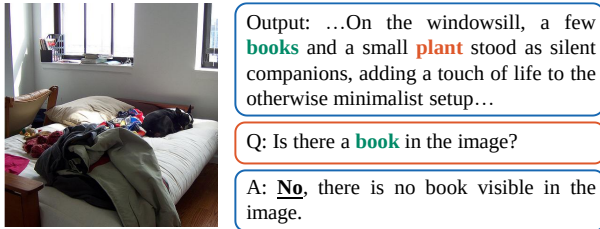


Figure 6: An example of hallucination in voluntary imagination. The model generates an output that includes “books” (which are actually present in the image), but in the subsequent presence evaluation, it incorrectly judges them as absent. This reflects a grounding failure regarding the generated objects.

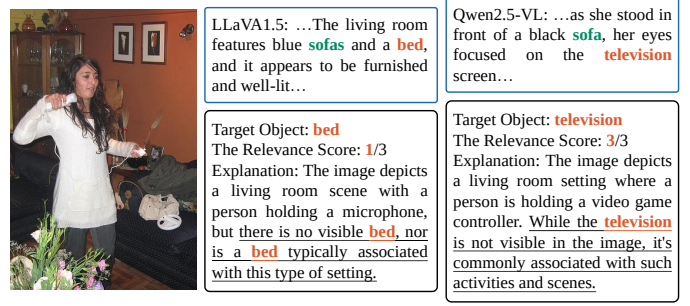


Figure 7: Examples of output objects with low and high relevance scores. Objects that are unlikely to appear in the scene of a given image tend to receive lower scores, e.g., a “bed” in a living room scene. Conversely, objects that are likely to co-occur in the same scene receive higher scores, e.g., a “television” in the same scene.

imagination and hallucination. When generating an imagined object (e.g., “people”), the model assigns relatively low attention to the image, relying mainly on previously generated textual context for imagination. In contrast, when generating a hallucinated object (e.g., “bowl”), the model allocates excessive attention to an incorrect region of the image, mistakenly identifying another object as a “bowl”. This suggests that although the model allocates more attention to the image tokens, hallucinations can still occur because it attends to incorrect regions.

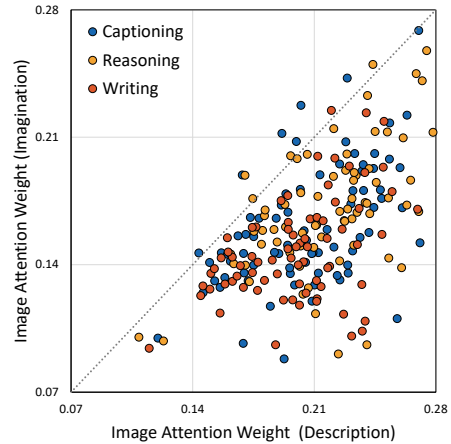


Figure 8: LLaVA1.5’s image attention weights when generating factual description and voluntary imagination objects. The model tends to allocate more image attention weight during factual description, while assigning less image attention weight during voluntary imagination.

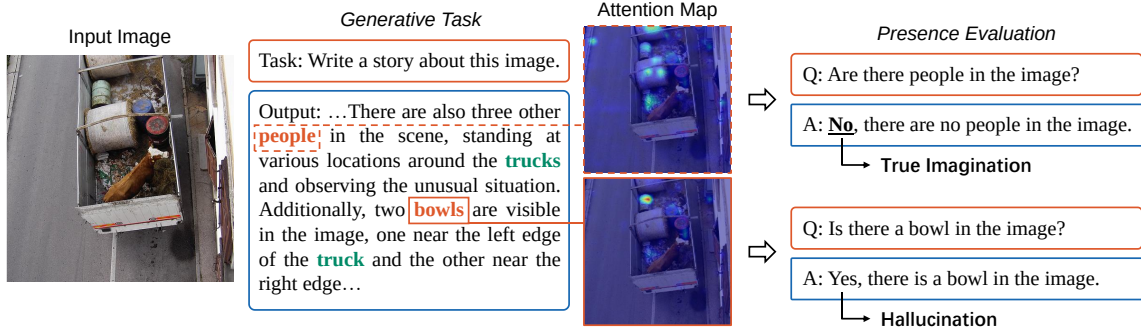


Figure 9: Visualization of LLaVA1.5’s attention maps when generating different objects. In the case of true imagination, the model allocates relatively low attention to the image. However, when hallucinating within the factual description, the model focuses its attention on a particular, erroneous visual region.

Prompt:

You will be provided with an image and an object name. Your task is to evaluate the relevance between the given object and the content of the image, and assign a score based on the following criteria:

Score 3, Highly Relevant : The object is commonly associated with or co-occurs with objects in the image, or typically appears in the scene depicted.

Score 2, Moderately Relevant: The object does not necessarily appear in such a scene, but its presence would not be inconsistent or unnatural.

Score 1, Irrelevant: The object is unlikely to appear in the type of scene shown in the image.

Please analyze the image and the object, then provide:

A. The relevance score (1-3)

B. A brief explanation in English supporting your score

Object: <Target Object>

Image: <Image Tokens>

Figure 10: The prompt used for relevance evaluation.

Table 5: Results of LVLMs’ hallucination rates under different expressive tendencies. The hyperparameter α is used to control the model’s expressive tendency.

α	LLaVA1.5			Gemma3			Qwen2.5-VL		
	Hal-D ↓	Hal-I ↓	Exp	Hal-D ↓	Hal-I ↓	Exp	Hal-D ↓	Hal-I ↓	Exp
-1	21.5 -0.9	14.9 +1.3	10.1 -2.1	15.3 +0.2	16.5 -2.0	18.5 -6.3	7.5 +0.2	41.0 +5.1	16.2 -14.1
-0.5	21.8 -0.5	14.5 +1.0	11.1 -1.0	15.6 +0.5	17.8 -0.7	20.7 -4.1	7.2 +0.0	41.4 +5.6	17.1 -13.1
0	22.3 +0.0	13.6 +0.0	12.1 +0.0	15.1 +0.0	18.5 +0.0	24.8 +0.0	7.3 +0.0	35.8 +0.0	30.2 +0.0
0.5	22.2 -0.2	13.3 -0.2	13.9 +1.8	14.6 -0.5	17.4 -1.1	27.0 +2.3	7.2 -0.1	36.5 +0.7	36.5 +6.3
1	21.9 -0.5	12.4 -1.1	14.2 +2.0	14.6 -0.5	18.4 -0.1	27.2 +2.4	7.1 -0.2	36.0 +0.2	37.5 +7.3

C Prompt for Relevance Assessment

The prompt used for relevance evaluation is shown in Fig. 10. We provide the MLLM judge with the criteria for determining different relevance scores and require it to include an explanation after assigning the relevance score.

D Contrastive Decoding for Expression Control

In this section, we provide the implementation details and additional results for our expression-control analysis. Specifically, we employ contrastive decoding to investigate whether explicitly manipulating a model’s expressive tendency (measured by *Exp*) impacts its underlying hallucination rates (*Hal-D* and *Hal-I*).

Contrastive decoding typically operates by contrasting the output probability distributions under different input conditions. Building upon the formulation in VCD [23], we contrast the model’s logits under a voluntary imagination prompt (e.g., story writing) against those from a factual description prompt (e.g., image captioning). This difference serves as a guiding signal to explicitly modulate the model’s propensity to imagine. The modified probability distribution for generating a new token is computed as follows:

$$p_{cd}(y|v, x_w, x_c) = \text{softmax}[\text{logit}_\theta(y|v, x_w) + \alpha(\text{logit}_\theta(y|v, x_w) - \text{logit}_\theta(y|v, x_c))], \quad (4)$$

where v is the visual input, x_w and x_c denote the prompts for the writing and captioning tasks respectively, y is the target token, and logit_θ represents the original model logits.

The hyperparameter α controls both the direction and magnitude of the contrastive adjustment. When $\alpha < 0$, the output distribution shifts toward the factual captioning task, resulting in a decreased expressive tendency. Conversely, when $\alpha > 0$, the distribution is pushed further away from the factual baseline, encouraging a higher expressive tendency.

As demonstrated in Table 5, varying α successfully and substantially alters the models’ expressive tendency *Exp*. However, despite these significant shifts in generation style, both the factual (*Hal-D*) and imagination-based (*Hal-I*) hallucination rates remain notably stable. This finding is critical: it demonstrates that the severe hallucination behavior observed in voluntary imagination is not simply a byproduct of the model “talking too much” or being overly expressive. Instead, it reaffirms our main-text conclusion that voluntary imagination suffers from a fundamental deficiency in internal visual grounding, which cannot be resolved by merely adjusting response-level expression.

E Complete Results for Relevance Assessment

The complete relevance assessment results obtained from the three generative experiments are shown in Table 6, Table 7 and Table 8. In the tables, D_H and I_T indicate whether the evaluated object belongs to a hallucination or a true imagination. **Score 1**, **Score 2**, and **Score 3** represent objects that are completely irrelevant, moderately relevant, and highly relevant to the image content, respectively. The numbers in the tables represent the count of objects that received the corresponding relevance score.

Table 6: Results of relevance assessment in the captioning task.

Model	Category	Score 1	Score 2	Score 3
LLaVA1.5 [27]	D_H	324	1041	2040
	I_T	407	821	809
LLaMA3.2-Vision [14]	D_H	234	342	1281
	I_T	87	115	187
InternVL2.5 [5]	D_H	81	208	1022
	I_T	288	305	429
Qwen2.5-VL [45]	D_H	30	49	731
	I_T	208	318	595
MiniCPM-o 2.6 [56]	D_H	56	166	861
	I_T	141	230	452
Gemma3 [43]	D_H	138	245	1171
	I_T	205	187	230

Table 7: Results of relevance assessment in the reasoning task.

Model	Category	Score 1	Score 2	Score 3
LLaVA1.5 [27]	D_H	204	588	1562
	I_T	289	503	877
LLaMA3.2-Vision [14]	D_H	162	236	1125
	I_T	85	127	331
InternVL2.5 [5]	D_H	33	74	523
	I_T	84	194	525
Qwen2.5-VL [45]	D_H	9	31	465
	I_T	269	482	1223
MiniCPM-o 2.6 [56]	D_H	26	86	584
	I_T	100	232	583
Gemma3 [43]	D_H	94	207	1025
	I_T	128	182	490

Table 8: Results of relevance assessment in the writing task.

Model	Category	Score 1	Score 2	Score 3
LLaVA1.5 [27]	D_H	345	1135	2147
	I_T	467	875	1003
LLaMA3.2-Vision [14]	D_H	207	312	1293
	I_T	117	196	379
InternVL2.5 [5]	D_H	53	199	848
	I_T	729	838	897
Qwen2.5-VL [45]	D_H	16	69	654
	I_T	584	773	1473
MiniCPM-o 2.6 [56]	D_H	53	153	870
	I_T	397	493	764
Gemma3 [43]	D_H	153	295	787
	I_T	760	703	659